

incompressible, acoustic compact ($\lambda \gg r$) $\quad \hat{q} = A\hat{u}$

Rayleigh conductivity $\quad K_R = i\omega\rho \dfrac{\hat{q}}{\hat{p}_1 - \hat{p}_2}$

$$K_R = \frac{A}{l_{eff}} \qquad l_{eff} = l + 2l'$$

$$\frac{\pi r^2}{l + \frac{16}{3\pi} r} < K_R < \frac{\pi r^2}{l + \frac{\pi}{2} r}$$

Impedance modeling of perforations

dimension $\ll$ acoustic wavelength

$$Z = R + i\left(\omega M - \frac{1}{\omega C}\right)$$

R —— resistance

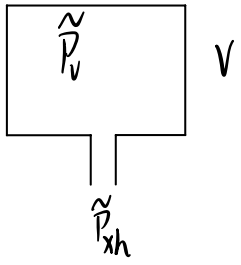
M —— intertance

C —— compliance

Application

Helmholtz Resonator

① Derivation 1



1° Rayleigh conductivity

$$\frac{i\omega \bar{\rho}_n \tilde{Q}_n}{\tilde{\bar{P}}_V - \tilde{\bar{P}}_{nh}} = K_R \quad \Rightarrow \quad \tilde{\bar{P}}_{nh} = \tilde{\bar{P}}_V - \frac{i\omega \bar{\rho}_n \tilde{Q}_n}{K_R}$$

$$= \tilde{\bar{P}}_V - \frac{i\omega \bar{\rho}_n S_n \tilde{u}_n}{K_R}$$

2° isentropic

$$\frac{P_V'}{\bar{c}_V^2} = \rho_V'$$

3° conservation of mass

$$\frac{\partial \rho_V'}{\partial t} V + \bar{\rho}_n S_n u_n' = 0 \xrightarrow{FT} i\omega \tilde{\bar{\rho}}_V V + \bar{\rho}_n S_n \tilde{u}_n = 0$$

$$\Rightarrow \quad \tilde{\bar{\rho}}_V = - \frac{\bar{\rho}_n S_n \tilde{u}_n}{i\omega V}$$

Combine these equations

$$\Rightarrow \quad \tilde{\bar{P}}_{nh} = - \frac{\bar{c}_V^2 \bar{\rho}_n S_n \tilde{u}_n}{i\omega V} - \frac{i\omega \bar{\rho}_n S_n \tilde{u}_n}{K_R}$$

$$= -\left(\frac{\bar{C}_V^2}{i\omega V} + \frac{i\omega}{K_R}\right)\bar{\rho}_n S_n \tilde{u}_n$$

$$\Rightarrow \quad Z_{HR} = \frac{\tilde{p}_{xh}}{-\tilde{u}_n} = \left(\frac{i\omega}{K_R} + \frac{\bar{C}_V^2}{i\omega V}\right)\bar{\rho}_n S_n$$

And $m_n = \rho_n S_n u_n \Rightarrow \tilde{m}_n = \bar{\rho}_n S_n \tilde{u}_n + S_n \bar{u}_n \tilde{\rho}_n$ ↘ $\tilde{\rho}_n = \tilde{\rho}_V$

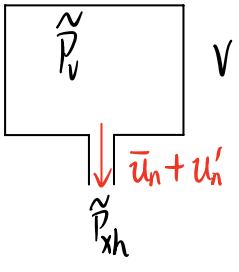
$$= \bar{\rho}_n S_n \tilde{u}_n + S_n \bar{u}_n \tilde{\rho}_V$$

$$= \bar{\rho}_n S_n \tilde{u}_n - \frac{\bar{\rho}_n S_n^2 \bar{u}_n \tilde{u}_n}{i\omega V}$$

$$= \left(1 - \frac{S_n \bar{u}_n}{i\omega V}\right)\bar{\rho}_n S_n \tilde{u}_n$$

$$= \left(1 - \frac{S_n \bar{u}_n}{i\omega V}\right)\bar{\rho}_n S_n \frac{\tilde{p}_{xh}}{Z_{HR}}$$

② Derivation 2



1° Rayleigh conductivity

$$\frac{i\omega\bar{\rho}_n \tilde{Q}_n}{\tilde{p}_V - \tilde{p}_{xh}} = K_R \quad \Rightarrow \quad \tilde{p}_{xh} = \tilde{p}_V - \frac{i\omega\bar{\rho}_n \tilde{Q}_n}{K_R}$$

$$= \tilde{P}_V - \frac{i\omega \bar{\rho}_n S_n \tilde{u}_n}{K_R}$$

2° isentropic

$$\frac{P_V'}{\bar{c}_V^2} = \rho_V'$$

3° conservation of mass

$$\frac{\partial \rho_V'}{\partial t} V + \bar{\rho}_n S_n u_n' + S_n \bar{u}_n \rho_n' = 0$$

$$\xrightarrow{FT} \quad i\omega \tilde{\rho}_V V + \bar{\rho}_n S_n \tilde{u}_n + S_n \bar{u}_n \tilde{\rho}_n = 0$$

$$\downarrow \quad \tilde{\rho}_n = \tilde{\rho}_V$$

$$\tilde{\rho}_V = - \frac{\bar{\rho}_n S_n \tilde{u}_n}{i\omega V + S_n \bar{u}_n}$$

Combine these equations,

$$\Rightarrow \quad \tilde{P}_{sh} = - \frac{\bar{c}_V^2 \bar{\rho}_n S_n \tilde{u}_n}{i\omega V + S_n \bar{u}_n} - \frac{i\omega \bar{\rho}_n S_n \tilde{u}_n}{K_R}$$

$$= - \left(\frac{\bar{c}_V^2}{i\omega V + S_n \bar{u}_n} + \frac{i\omega}{K_R} \right) \bar{\rho}_n S_n \tilde{u}_n$$

$$\Rightarrow \quad Z_{HR} = \frac{\tilde{P}_{sh}}{-\tilde{u}_n} = \left(\frac{\bar{c}_V^2}{i\omega V + S_n \bar{u}_n} + \frac{i\omega}{K_R} \right) \bar{\rho}_n S_n$$

And $m_n = \rho_n S_n u_n \Rightarrow \tilde{m}_n = S_n \bar{\rho}_n \tilde{u}_n + S_n \bar{u}_n \tilde{\rho}_n$

$\tilde{\rho}_n = \tilde{\rho}_v$

$$= S_n \bar{\rho}_n \tilde{u}_n + S_n \bar{u}_n \tilde{\rho}_v$$

$$= S_n \bar{\rho}_n \tilde{u}_n - \frac{S_n^2 \bar{\rho}_n \bar{u}_n \tilde{u}_n}{i\omega V + S_n \bar{u}_n}$$

$$= \left(1 - \frac{S_n \bar{u}_n}{i\omega V + S_n \bar{u}_n}\right) \bar{\rho}_n S_n \tilde{u}_n$$

$$= \frac{i\omega V}{i\omega V + S_n \bar{u}_n} \bar{\rho}_n S_n \tilde{u}_n$$

$$= \frac{-1}{1 + \frac{S_n \bar{u}_n}{i\omega V}} \frac{\bar{\rho}_n S_n}{Z_{HR}} \tilde{p}_{xh}$$

③ Comparison with two derivation method.

$$\tilde{m}_n = \frac{1}{1 + \frac{S_n \bar{u}_n}{i\omega V}} \bar{\rho}_n S_n \tilde{u}_n$$

If $\left|\frac{S_n \bar{u}_n}{i\omega V}\right| << 1$, $\frac{1}{1+x} = 1 - x + x^2 + \cdots$

$$\Rightarrow \tilde{m}_n \approx \left(1 - \frac{S_n \bar{u}_n}{i\omega V}\right) \bar{\rho}_n S_n \tilde{u}_n$$

$\Rightarrow$ We can conclude that the first method

is only correct only when $\left|\frac{S_n \bar{u}_n}{i\omega V}\right| \ll 1$,

④ The classical Helmholtz model

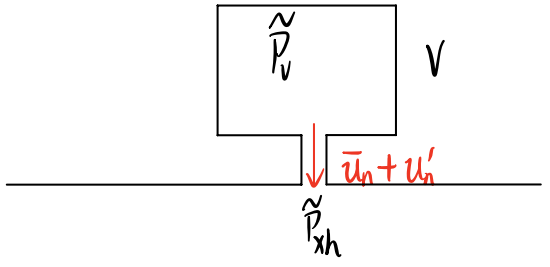
1° The hole model:

$$\tilde{p}_v - \tilde{p}_{xh} = i\omega l_{eff} \bar{\rho}_n \tilde{u}_n + C_d \bar{\rho}_n \bar{u}_n \tilde{u}_n$$

$$\Rightarrow \quad \tilde{p}_{xh} = \tilde{p}_v - \left(i\omega \bar{\rho}_n l_{eff} + C_d \bar{\rho}_n \bar{u}_n\right) \tilde{u}_n$$

2° isentropic

$$\frac{\tilde{p}_v}{\bar{c}_v^2} = \tilde{\rho}_v$$



3° conservation of mass

$$i\omega \tilde{\rho}_v V + \bar{\rho}_n S_n \tilde{u}_n = 0$$

$$\Rightarrow \quad \tilde{\rho}_v = -\frac{\bar{\rho}_n S_n \tilde{u}_n}{i\omega V}$$

Combine these equations,

$$\Rightarrow \quad \tilde{p}_{xh} = -\frac{\bar{c}_v^2 \bar{\rho}_n S_n \tilde{u}_n}{i\omega V} - \left(i\omega \bar{\rho}_n l_{eff} + C_d \bar{\rho}_n \bar{u}_n\right) \tilde{u}_n$$

$$= -\left(\frac{\bar{c}_v^2 \bar{\rho}_n S_n}{i\omega V} + i\omega \bar{\rho}_n l_{eff} + C_d \bar{\rho}_n \bar{u}_n\right) \tilde{u}_n$$

$$\Rightarrow \quad Z_{HR} = \frac{\tilde{p}_{xh}}{-\tilde{u}_n} = \bar{\rho}_n \left(\frac{S_n \bar{C}_V^2}{i\omega V} + i\omega l_{eff} + C_d \bar{u}_n \right)$$

$$= \bar{\rho}_n \left[i\omega \left(l_{eff} - \frac{S_n \bar{C}_V^2}{\omega^2 V} \right) + C_d \bar{u}_n \right]$$

W_{re}

And $\quad \frac{Z_{HR}}{\bar{\rho}_n \bar{C}_n} = r + iX$

r: resistance 声阻

X: reactance 声抗